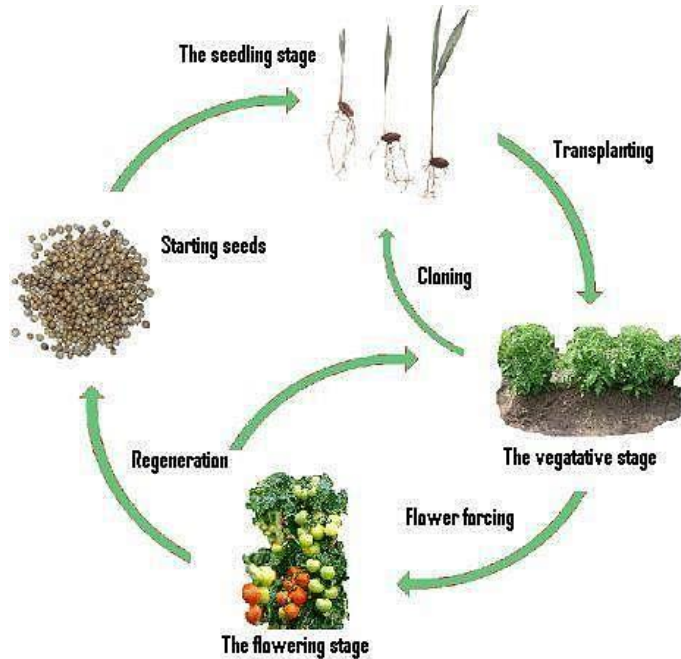


Plant Life (Part - 5)

PLANT LIFE



FUNDAMENTAL

- Paddy, wheat and maize are monocot plants, belong to grass family.
- Pea, gram and other pulses are dicot plants and belong to leguminous family.
- Monocot plants have one cotyledon whereas dicot plants have two cotyledons in their seeds.
- Gymnosperms are group of plants bear which naked seeds, e.g., Pinus, Cycas
- Angiosperm plants are highly developed flowering plants, e.g., Mango, Neem.
- Mushrooms are grown in dark places; it doesn't undergo photosynthesis rather it feeds on dead and decayed organic matter.

Types of plant on basis of habitat

1. Terrestrial
2. Aquatic plant
3. Epiphytes (Air plants)
4. Marshy plant

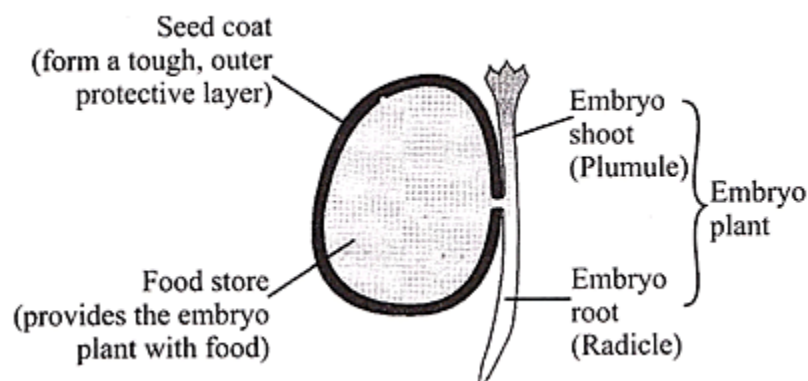
Types of Plants on the basis of habitat
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A.	Terrestrial Plants grow on the land e.g. mango, pine, coconut	1. Desert Plants: Leaves reduced into spine, swollen stem to store water, long roots etc. e.g. Cactus 2. Plants in hot and damp places remain evergreen and never shed leaves e.g. coconut, neem. 3. Plants of plains: They have branches and leaves shed leaves in autumn e.g. maple tree 4. Plants of hilly areas: Tall plants with lots of branches of flowers e.g. pine, cycas
B.	Aquatic Plants grow in water e.g. Lotus, Vallisneria	1. Floating Plants: Floats on the surface of pond, lake etc. e.g. water hyacinth, duckweed 2. Submerged Plants: Breathe through the body surface. Have no pores on the leaves e.g. Hydrilla, Vallisneria etc.
C.	Epiphytes (Air Plants) those live on trees e.g. mosses, orchids	1. Not permanently rooted in soil 2. Developed root system 3. Stem not extensively developed 4. Reduction in leaf number
D.	Marshy Plants have breathing roots to take air from atmosphere e.g. mangroves	1. Marshy plants are characterized by luxuriant growth and often have large leaves. 2. Close to the water's edge, there is no shortage of water for growth. 3. Plants in this region can therefore afford to have large leaves since excessive loss of moisture from the foliage is not a problem.

Adaptation in Different Plants	
Valisneria	Submerged plants have pores on the leaf surface to breathe
Water Hyacinth	Floating plants have long stem and broad leaves that are buoyant
Desert Plants	Desert plants have leaves reduced to spines to prevent water loss through transpiration. Long roots to go up to underground water sources

Drosera	Insectivorous plants are attractive and colorful to attract insect and birds
Lotus	Fixed plants have long hollow stem to provide support and broad leaves to provide buoyancy
Plants growing in hilly area	Plants growing in hilly area have leaves reduced to needle shaped to allow snow to slip off the leaf surface
Marshy plant	Marshy plants have air breathing roots to absorb nutrients from the air

A seed is a part of the fruit which gives birth to a new plant. It is surrounded and protected by seed coat. A seed requires water, oxygen and the optimum temperature to germinate.



Seed dispersal is the phenomenon of carrying the fruits or seeds to distant area in order to spread them.

Fruits and seeds are dispersed so that young plants can get sufficient space, light, water and nutrients to grow well.

Method of seed dispersal	Characteristics	
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By Wind	The fruits or seeds are usually small and light or have large surface area so that they can be carried easily by the wind e.g., Cotton, poppy, maple, drumsticks	
By Water	The fruits or seeds usually have a fibrous structure that to float on water. E.g., coconut, lotus, water lily and hydrilla	
By Animals	The fruits which are sweet and juicy are eaten by animals. The seeds are usually not digested are dropped out e.g., Guava, Mango, Berries	
By Explosion	The fruits dry up and split open or release or scatter the seeds inside. E.g., Balsam, Pea, Geranium	

REPRODUCTION IN PLANT

- **Sexual Reproduction** - Flower is the main reproductive organ
- **Asexual Reproduction** - Vegetative propagation

1. Vegetative reproduction by Stem

- **Runner**- In strawberries their stems run horizontally parallel to the ground. New roots emerge from the nodes and gives rise to new plants.
- **Suckers** - Similar to runners but have shorter internodes. e.g., Mint, Chrysanthemum.
- **Rhizome** - It has underground node and internodes. e.g., Ginger
- **Tuber** - Vegetative production through bud eyes. e.g., Potato
- **Corm** - Is condensed rhizome which is formed vertically in the ground, e.g., Choco, rice.
- **Bulbs** - In non-favorable situation, bulb in the plants grows into plants, e.g., Onion.
- **Vegetative propagation by Leaf**
- Buds present on the margin of leaves develops into the new plants on getting the favorable conditions.

Vegetative propagation by Root

- Tuberous roots of sweet potato and dahlia when get detached from the plant, grows into new plant.

Artificial means of Vegetative propagation

1. **Cutting**:- Cutting in rose, bougainvillea, sugarcane etc. can grow into new plants when put in soil.
2. **Grafting**:- It is the technique in which stem of one desired plant is inserted on the root of other desired plants, e.g., Rose, mango, apple, etc. (only applicable for dicot plants).
3. **Layering**:- When bark, removed from a branch of a plant, is bent down and touches moist soil growing upright, e.g. Jasmine, Strawberries and Grapevine.

CROPS

Plants of one kind grown in a particular area or region during a particular period are called crops.

Kharif crops:- (Rice, maize, jowar and bajra) grown in summer season.

Rabi crops:- (Wheat, barley, gram) grown in winter season.

Oil producing crops:- Sunflower, Groundnut, Coconut, Mustard and Peanuts.

Fiber producing crops:- Cotton, Jute

Cash crops:- Which are raw materials for industries. (Rubber, Tea, Coffee, Sugarcane)

Food crops:- For food materials (Paddy, wheat)

Medicinal Plants:- Uses for herbal medicine. (Aloevera, Ashwagandha, Margosa, Wood apple)